

Tutorial Unit-II

①.

x	y	X	Y	XY	X^2	Y^2
36	29	6	4	24	36	16
23	18	-7	-7	49	49	49
27	20	-3	-5	15	9	25
28	22	-2	-3	6	4	9
29	27	-2	2	-4	4	4
29	21	-1	-4	4	1	16
30	29	0	4	0	0	16
31	27	1	2	2	1	4
33	29	3	4	12	9	16
35	28	5	3	15	25	9

$$\Sigma x = 300 \quad \Sigma y = 250$$

$$\Sigma XY = 123 \quad \Sigma X^2 = 138 \quad \Sigma Y^2 = 164$$

$$\bar{x} = \frac{\Sigma x}{n} = 30 ; \quad \bar{y} = \frac{\Sigma y}{n} = 25$$

$$\sigma_x^2 = \frac{n}{n} \Sigma x^2 = 138 ; \quad \sigma_y^2 = \frac{\Sigma y^2}{n} = 16.4$$

$$\sigma_x = 3.7148 ; \quad \sigma_y = 4.0497$$

$$r = \frac{\Sigma XY}{\sqrt{\Sigma x^2 \Sigma y^2}} = \frac{123}{\sqrt{138 \cdot 164}} = 0.8176$$

$$(y - \bar{y}) = r \frac{\sigma_y}{\sigma_x} (x - \bar{x}) \quad ; \quad (x - \bar{x}) = r \frac{\sigma_x}{\sigma_y} (y - \bar{y})$$

$$y - 25 = 0.8176 \left(\frac{4.0497}{3.7148} \right) (x - 30) \quad ; \quad (x - 30) = 0.8176 \left(\frac{3.7148}{4.0497} \right) (y - 25)$$

$$y = 0.893x - 1.7393 \quad ; \quad x = 0.75y + 11.2503$$

②.

Given $\bar{x} = 40 ; \bar{y} = 60$
 $\sigma_x = 10 ; \sigma_y = 15$
 $r = 0.7$

$$\Rightarrow (y - \bar{y}) = r \frac{\sigma_y}{\sigma_x} (x - \bar{x})$$

$$\Rightarrow (y - 60) = 0.7 \left(\frac{15}{10} \right) (x - 40)$$

$$\Rightarrow y = 1.05x + 18$$

③

Given $x - y = 0$ — ①
 $4x - y - 3 = 0$ — ②

i) Wkt $\bar{x}$ & $\bar{y}$ lie on these lines,

ii) Solving ① & ②,

$$\left. \begin{array}{l} \bar{x} - \bar{y} = 0 \\ 4\bar{x} - \bar{y} - 3 = 0 \end{array} \right\} \quad \bar{x} = 1, \bar{y} = 1$$

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iii) Consider $x - y = 0$

$$\Rightarrow y = x$$

$$\Rightarrow b_{yx} = 1$$

Now $1x - y - 3 = 0$

$$\Rightarrow x = \frac{y}{1} + \frac{3}{1}$$

$$b_{xy} = 0.25$$

$$r = \sqrt{b_{xy} b_{yx}} = \sqrt{0.25} = 0.5$$

iv) $r \frac{\sigma_y}{\sigma_x} = 1$

$$\Rightarrow \sigma_y = \frac{\sigma_x}{r} = \frac{1}{0.5} = 2$$

10/12
↑

Given

$$\bar{x} = 53, \bar{y} = 14.2$$

$$\sigma_x^2 = 130, \sigma_y^2 = 165$$

$$\sigma_x = 11.4018, \sigma_y = 12.8452$$

$$\sum XY = 1220$$

$$\sum X^2 = \sigma_x^2 n = 130 \times 10 = 1300$$

$$\sum Y^2 = \sigma_y^2 n = 165 \times 10 = 1650$$

$$= 0.9385(45) + 92.262$$

⑤

$$r = \frac{\sum XY}{\sqrt{\sum X^2} \sqrt{\sum Y^2}} = \frac{1220}{\sqrt{1300} \sqrt{1650}} = 0.833$$

$$y - \bar{y} = r \frac{\sigma_y}{\sigma_x} (x - \bar{x})$$

$$y - 14.2 = 0.833 \left(\frac{12.8452}{11.4018} \right) (x - 53)$$

$$y = 0.9385x + 92.262$$

$$x - \bar{x} = r \frac{\sigma_x}{\sigma_y} (y - \bar{y})$$

$$x - 53 = 0.833 \left(\frac{11.4018}{12.8452} \right) (y - 14.2)$$

$$x = 0.7394y - 51.9948$$

$$r_{23.1} = \frac{r_{23} - r_{12} r_{13}}{\sqrt{(1 - r_{12}^2)(1 - r_{13}^2)}} = \frac{0.5 - 0.4(0.6)}{\sqrt{(1 - 0.4^2)(1 - 0.6^2)}} = 0.3546$$

$$r_{1.23} = \frac{r_{12}^2 + r_{13}^2 - 2r_{12}r_{13}r_{23}}{1 - r_{23}^2} = \frac{0.4^2 + 0.6^2 - 2(0.4)(0.6)(0.5)}{1 - 0.5^2}$$

$$= 0.6110$$

⑥

y	x ₁	x ₂	x ₃	x ₁ x ₂	x ₁ x ₃	x ₂ x ₃	x ₁ ²	x ₂ ²	x ₃ ²
905	85	75	78	6375	5850	6630	7225	5625	6084
850	98	56	56	5488	3136	5488	9604	3136	3136
910	97	42	80	4074	3360	7260	9409	1764	6400
1050	85	45	64	3825	2880	5440	7225	2025	4096
Σ	365	218	278	19762	15246	15318	33444	13550	17716

$$\bar{x}_1 = 91.25; \bar{x}_2 = 54.5; \bar{x}_3 = 69.5$$

$$r_{12} = \frac{\sum x_1 x_2 - n \bar{x}_1 \bar{x}_2}{\sqrt{(\sum x_1^2 - n \bar{x}_1^2)(\sum x_2^2 - n \bar{x}_2^2)}} = -0.0084$$

$$r_{13} = \frac{\sum x_1 x_3 - n \bar{x}_1 \bar{x}_3}{\sqrt{(\sum x_1^2 - n \bar{x}_1^2)(\sum x_3^2 - n \bar{x}_3^2)}} = -0.0026$$

$$r_{23} = \frac{\sum x_2 x_3 - n \bar{x}_2 \bar{x}_3}{\sqrt{(\sum x_2^2 - n \bar{x}_2^2)(\sum x_3^2 - n \bar{x}_3^2)}} = 0.0063$$

$$R = \frac{1}{1.23} \sqrt{\frac{r_{12}^2 + r_{13}^2 - 2r_{12}r_{13}r_{23}}{1 - r_{23}^2}} = 0.0088$$

(7) $y = a + b_1 x_1 + b_2 x_2$

NE: $\sum y = 6a + b_1 \sum x_1 + b_2 \sum x_2$

$$\sum x_1 y = a \sum x_1 + b_1 \sum x_1^2 + b_2 \sum x_1 x_2$$

$$\sum x_2 y = a \sum x_2 + b_1 \sum x_1 x_2 + b_2 \sum x_2^2$$

y	x_1	x_2	$x_1 x_2$	$x_1 y$	$x_2 y$	x_1^2	x_2^2
160	4	1.5	6	640	240	16	2.25
112	3	2.2	6.6	336	246.4	9	4.84
69	1.6	1	1.6	110.4	69	2.56	1
90	1.2	2	2.4	72	180	1.44	4
123	3.4	0.8	2.72	418.2	98.4	11.56	0.64
186	4.8	1.6	7.68	892.8	297.6	23.04	2.56
Σ 740	18	9.1	27	2469.4	1131.4	63.6	15.29

$$\Rightarrow 240 = 6a + 18b_1 + 9.1b_2$$

$$\Rightarrow 2469.4 = 18a + 63.6b_1 + 27b_2$$

$$\Rightarrow 1131.4 = 9.1a + 2.2b_1 + 15.29b_2$$

$$\therefore a = 27.0368; b_1 = 26.3354; b_2 = 11.4002$$

$$\therefore y = 27.0368 + 26.3354x_1 + 11.4002x_2$$

$$y(3.00, 2.60) = 27.0368 + 26.3354(3.00) + 11.4002(2.60) = 154.1183$$

⑧ NOT IN QB

Q

y	x_1	x_2	$x_1 x_2$	x_1^2	x_2^2	$x_1 y$	$x_2 y$
10	0	100	0	0	10000	0	1000
48	70	100	7000	4900	10000	3366	4800
41	35	140	4900	1225	19600	1435	5740
40	0	110	0	0	32400	0	2200
39	70	110	12600	4900	32400	2230	7020
44	70	140	9800	4900	19600	3080	6160
24	0	140	0	0	19600	0	3360
31	35	100	3500	1225	10000	1085	3100
44	35	180	6300	1225	32400	1546	7920
Σ	321	315	1260	66100	18375	18600	13230

$$\text{NE's: } \Sigma y = 9\beta_0 + \beta_1 \Sigma x_1 + \beta_2 \Sigma x_2$$

$$\Sigma x_1 y = \beta_0 \Sigma x_1 + \beta_1 \Sigma x_1^2 + \beta_2 \Sigma (x_1 x_2)$$

$$\Sigma x_2 y = \beta_0 \Sigma x_2 + \beta_1 \Sigma (x_1 x_2) + \beta_2 \Sigma x_2^2$$

Substituting,

$$321 = 9\beta_0 + 315\beta_1 + 1260\beta_2$$

$$13230 = 315\beta_0 + 18375\beta_1 + 66100\beta_2$$

$$76300 = 1260\beta_0 + 66100\beta_1 + 18375\beta_2$$

$$\beta_0 = 6.3333, \beta_1 = 0.2214, \beta_2 = 0.1417$$

$$y = 6.3333 + 0.2214x_1 + 0.1417x_2$$

$$y_{(40,130)} = 6.3333 + 0.2214(40) + 0.1417(130) = 35.6103$$